Addendum to Version 1: Clarifications, Caveats, and Recommended Independent Replication

Addendum to: Nguyen, D. (2026). "The Radial Address of the NFW Cusp Failure: Stacked Rotation Curve Residuals Across 149 SPARC Galaxies." Zenodo. doi:10.5281/zenodo.20076361

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Purpose

This addendum addresses seven items identified after publication of the parent paper: (A) prior art in the scale-invariant stacked-rotation-curve lineage; (B) explanation of the sample reduction from N = 149 to N = 128 in the per-galaxy test; (C) beam smearing as an unmodeled systematic; (D) radial stellar mass-to-light ratio gradients as a potential bias; (E) uncertainty propagation from the photometric disk scale length r_d; (F) a stronger profile control to break the Burkert tautology; (G) explicit recommendations for independent replication on THINGS, LITTLE THINGS, and MIGHTEE-HI; and (H) a brief summary of which conclusions of the parent paper survive these clarifications and which require qualification. The principal empirical result of the parent paper — that the NFW fractional residual peaks negatively at $r/r_d \approx 0.6$ with per-galaxy median –0.099 and Wilcoxon $p = 4 \times 10^{-13}$ on N = 128 galaxies — is unaffected by the items below, but its interpretation is sharpened.

A. Prior Art: The Scale-Invariant Stacked Rotation Curve Lineage

Version 1 cited Steffen (2025) as the closest prior work on stacked SPARC analyses. A more complete account of prior art is warranted. The practice of stacking rotation curves in scale-normalized radial coordinates predates the present work by three decades and is foundational to the methodology used here.

Persic, Salucci & Stel (1996, MNRAS 281, 27) introduced the "universal rotation curve" by stacking rotation curves of 1100 spiral galaxies in coordinates normalized to the optical radius R_{opt}, demonstrating that scale-invariant stacking reveals systematic features that are obscured in physical-unit analyses. **Salucci & Burkert (2000, ApJ 537, L9)** refined this with the inclusion of low-surface-brightness systems. **Lelli, Fraternali & Verheijen (2013, MNRAS 433, L30)** — using the same SPARC team that produced the dataset of the present work — applied scale-invariant stacking to derive a universal velocity profile. **Karukes & Salucci (2017, MNRAS 465, 4703)** applied the same family of methods to dwarf irregulars.

The contribution of the parent paper is therefore best characterized as: applying the Persic-Salucci-Lelli scale-invariant stacking methodology to *fractional residuals from parametric profile fits* (rather than to the rotation curves themselves), and quantifying per-galaxy significance for the resulting feature across three dark matter density profiles. This is an incremental methodological refinement on a 30-year-old tradition, not a new technique. The principal new content is the specific scaled radius ($r/r_d \approx 0.6$) and its per-galaxy significance ($p = 4 \times 10^{-13}$) — both of which are unaffected by this re-attribution.

B. Sample Reduction: N = 149 to N = 128 in the Per-Galaxy Test

Section 2 of the parent paper specifies N = 149 galaxies meeting the SPARC quality and inclination cuts. Section 4.2 reports the per-galaxy Wilcoxon test on N = 128 galaxies without explaining the 21-galaxy reduction. The omission is corrected here.

The per-galaxy test requires each galaxy to contribute at least one measured radial point within the analysis zone $0.4 < r/r_d < 0.8$. Twenty-one of the 149 galaxies meeting the global quality criteria have no rotation curve sampling in this scaled-radius interval. These galaxies fall into three categories: (i) galaxies whose innermost observed radius exceeds 0.8 r_d (typically large, well-resolved spirals where the first H I beam is at large physical radius); (ii) galaxies with sparse inner sampling such that no point lies in [0.4, 0.8] r_d; and (iii) one galaxy where the photometric r_d is so small that the analysis zone falls entirely below the H I beam size. The 21 excluded galaxies are not preferentially of any morphological type or mass class; the exclusion is driven by radial sampling, not selection.

The 128-galaxy per-galaxy result and the 149-galaxy stacked result are therefore consistent: the stacked analysis uses all 149 galaxies' data points that happen to fall in the zone, while the per-galaxy test uses 128 galaxies that contribute at least one such data point. The discrepancy between the numbers in the title and the per-galaxy test is bookkeeping, not selection bias.

C. Beam Smearing as an Unmodeled Systematic

The parent paper does not discuss beam smearing in the H I rotation curves of the SPARC sample. This is an oversight: at small physical radii (typically $r < 3$ kpc), finite beam size in interferometric H I observations artificially flattens the inner rotation curve, systematically reducing the measured V_{obs}. SPARC includes galaxy-specific beam-smearing corrections (Lelli, McGaugh & Schombert 2016, Section 4), but the residual uncertainty in these corrections is not propagated through the fractional residual calculation of the parent paper.

Because the NFW feature is located at small scaled radius ($r/r_d \approx 0.6$), the radial zone where beam smearing is most severe, the possibility that the feature is partially or wholly a beam-smearing artifact must be considered. Two arguments suggest this is unlikely to be the dominant cause but neither is dispositive:

First, the feature appears in galaxies spanning a wide range of physical sizes (r_d from 0.5 to > 5 kpc) and therefore a wide range of physical radii at fixed r/r_d. If beam smearing were the dominant cause, the feature should scale with physical (not scaled) radius, and stacking in r/r_d coordinates should wash it out. The fact that the feature appears at a specific *scaled* radius is evidence — though not proof — against a pure beam-smearing origin.

Second, the feature appears in Hα and CO rotation curves within SPARC (which have higher angular resolution than 21-cm) as well as in H I curves. A formal subsample comparison was not performed in the parent paper and is recommended as a follow-up: split the 128-galaxy sample into H I-only and high-resolution-tracer subsamples, repeat the analysis, and verify the feature is consistent across resolution classes. If the H I subsample shows the feature at 1.5 × the amplitude of the Hα/CO subsample, beam smearing is contributing materially. If the amplitudes are consistent, beam smearing is subdominant.

Until this subsample comparison is performed, the parent paper's identification of the NFW feature as a kinematic signature of the cusp's inner mass excess should be qualified as: "consistent with a cusp signature, with residual contribution from beam smearing not excluded at the per-galaxy level."

D. Radial Stellar Mass-to-Light Ratio Gradients

The parent paper assumes a galaxy-wide constant $\Upsilon_* = 0.5\ M_\odot/L_\odot$ at 3.6 μm (McGaugh & Schombert 2014) and tests sensitivity by varying the global value across $\Upsilon_* = 0.3, 0.5, 0.7, 0.8$. The feature survives these global variations. However, real galaxies exhibit *radial* $\Upsilon_*(r)$ gradients — typically older, more metal-rich stellar populations in the inner disk and younger, lower-Υ_* populations at larger radii. The global sensitivity test does not address this radial structure.

If, for example, the true inner Υ_* at $r/r_d \approx 0.5$ is 0.6 (older population) while the value assumed is 0.5 (population average), then V_{disk}^2 is under-subtracted from V_{obs}^2 in the inner region, leaving a residual

V_{DM}^2 that is artificially high. The NFW fit, which has the largest inner density of the three profiles, would then appear to over-predict — exactly the signal the parent paper reports. A radial $\Upsilon_*(r)$ gradient of order 20% from inner to outer disk could in principle produce a fractional residual of comparable magnitude to the observed feature.

Schombert et al. (2019, MNRAS 483, 1496) provide population-synthesis estimates of radial $\Upsilon_*(r)$ for SPARC galaxies. Adopting a radially varying $\Upsilon_*(r)$ calibrated to these estimates is the appropriate next-generation test. JWST near-infrared mapping (Lower et al. 2024; Williams et al. 2024) will provide galaxy-by-galaxy radial $\Upsilon_*(r)$ that can be substituted directly. Until this re-analysis is performed, the parent paper's claim that the feature is robust to mass-to-light ratio assumptions should be qualified to read: "robust to *global* Υ_* variations across the range 0.3-0.8; sensitivity to *radial* $\Upsilon_*(r)$ gradients has not been characterized and could in principle contribute to the observed amplitude."

E. Propagation of r_d Uncertainty

The disk scale length r_d used to construct the normalized radial coordinate is taken from SPARC's photometric decomposition at 3.6 μm. This decomposition is itself a fit, with statistical and systematic uncertainty. The parent paper treats r_d as exact when constructing the r/r_d coordinate. This is an idealization.

Statistical uncertainty on individual r_d values is typically 5-10% from SPARC's published error bars. Systematic uncertainty from the bulge-disk decomposition choice — exponential disk plus Sersic bulge, vs. free Sersic disk index, vs. multi-component models — adds an additional 5-15% spread for galaxies with bulges. Combined, individual r_d values have ~10-20% effective uncertainty. Propagated to the r/r_d coordinate, this smears the location of any genuine feature by approximately the same fraction.

A 15% smearing of the radial coordinate would broaden a delta-function feature into one of half-width $\Delta(r/r_d) \approx 0.1$ around the central location. The feature reported in the parent paper has half-width $\Delta(r/r_d) \approx 0.2$, comparable to but larger than the r_d-smearing scale. The peak location $r/r_d = 0.6 \pm 0.1$ reported in the parent paper's conclusions is therefore consistent with the resolution permitted by r_d systematics; a sharper underlying feature cannot be resolved without higher-fidelity photometric decompositions. The amplitude of the per-galaxy residual is not affected by r_d smearing, only the localization of the peak in scaled-radius space.

F. Stronger Profile Control: Breaking the Burkert Tautology

The parent paper uses the Burkert (1995) profile as a control: it has a flat core (no cusp), and shows no residual feature at $r/r_d \approx 0.6$ (median +0.006, p = 0.98). The parent paper interprets this as confirmation that the NFW feature is cusp-specific. This interpretation is correct in the limited sense that flat-core profiles do not produce the feature. However, the Burkert null result is partially tautological: any profile with a flat inner core will trivially fail to produce a cusp-driven overprediction, regardless of any other physical content.

A more demanding test uses a profile family with a tunable inner slope. The Zhao (1996) generalized αβγ family has density $\rho(r) = \rho_s / [(r/r_s)^\gamma (1 + (r/r_s)^\alpha)^{(\beta-\gamma)/\alpha}]$ with NFW recovered at $(\alpha, \beta, \gamma) = (1, 3, 1)$ and Burkert approximately recovered at $(\alpha, \beta, \gamma) \approx (2, 3, 0)$. A scan over $\gamma \in [0, 1]$ with fixed $(\alpha, \beta) = (1, 3)$ traces a continuous family of profiles whose inner slope varies from cored ($\gamma = 0$) to NFW-cusp ($\gamma = 1$). If the feature amplitude at $r/r_d \approx 0.6$ scales monotonically with γ, the cusp-specific interpretation is supported by a non-trivial control. If the feature amplitude shows other dependence (e.g., on β or on the inner concentration r_s), the cusp-specific interpretation requires qualification.

This scan has not been performed in the parent paper. It is recommended as a follow-up analysis. Until performed, the parent paper's claim of "cusp-specific feature" should be qualified to read: "feature is present in NFW (cuspy) but absent in Burkert (cored); a continuous-slope profile scan is recommended to confirm the cusp origin against alternative profile properties."

G. Recommended Independent Replication

The result of the parent paper is derived from a single dataset (SPARC) using a single set of photometric and kinematic measurements. Independent replication on alternative datasets is the strongest test of whether the feature is physical or a SPARC-specific systematic. Three datasets are recommended for follow-up:

THINGS (Walter et al. 2008, AJ 136, 2563): The H I Nearby Galaxy Survey provides high-resolution 21-cm rotation curves for ~30 nearby galaxies with better angular resolution than typical SPARC observations. A subset of THINGS galaxies overlap with SPARC, enabling direct cross-validation. If the feature appears in THINGS at consistent amplitude, beam smearing is sub-dominant; if it appears more strongly in SPARC than THINGS, beam smearing contributes materially.

LITTLE THINGS (Hunter et al. 2012, AJ 144, 134): Extension to dwarf irregular galaxies. The core-cusp problem is strongest in dwarfs (Oh et al. 2015). A LITTLE THINGS replication would test whether the $r/r_d \approx 0.6$ feature persists in the population where cusp-vs-core tension is most acute, or whether dwarfs show a different scaled-radius location for the feature.

MIGHTEE-HI (Maddox et al. 2021, A&A 646, A35; ongoing MeerKAT survey): Larger H I sample at slightly higher redshift (z ■ 0.5) with sub-arcsecond resolution at the survey's deep-tier pointings. A MIGHTEE-HI replication would extend the sample by an order of magnitude and provide the first test of whether the feature persists at slightly larger lookback time.

The recommended common analysis protocol is: (i) adopt the same three profile families (NFW, CIS or Burkert, with the addition of a Zhao $\alpha\beta\gamma$ scan per Section F); (ii) use the same r/r_d normalization; (iii) use the same fractional residual statistic; (iv) compute per-galaxy medians in the same $0.4 < r/r_d < 0.8$ zone; (v) report Wilcoxon p-values. A consistent result across THINGS, LITTLE THINGS, and MIGHTEE-HI would elevate the parent paper's finding from "feature in SPARC" to "feature in late-type spiral kinematics broadly." A divergent result would identify which dataset's systematics drive the SPARC feature.

H. Summary: What Stands and What Is Qualified

The principal empirical claim of the parent paper — that NFW fits to SPARC rotation curves produce a stacked fractional residual peaking negatively at $r/r_d \approx 0.6$ with per-galaxy median -0.099 and Wilcoxon $p = 4 \times 10^{-13}$ on N = 128 galaxies — is unchanged by the considerations above. The statistical result is real within the SPARC sample with SPARC photometry, SPARC kinematics, and the fitting methodology of the parent paper.

The following interpretive claims are **qualified**:

• "Robust to mass-to-light ratio assumptions" — robust to *global* Υ_* variations; sensitivity to radial $\Upsilon_*(r)$ gradients is uncharacterized (Section D).

• "Cusp-specific feature" — established against the Burkert flat-core profile; a continuous-slope profile scan is recommended to confirm against alternative profile properties (Section F).

• "$r/r_d \approx 0.6 \pm 0.1$" — consistent with the resolution permitted by r_d systematics; a sharper underlying feature would not be resolvable without improved photometric decomposition (Section E).

• "Kinematic signature of the cusp's inner mass excess" — consistent with a cusp signature, with residual contribution from beam smearing not formally excluded (Section C).

The forward-looking claim — that the stacked residual profile and per-galaxy detection rate can serve as a benchmark for hydrodynamical simulations of cusp-to-core transformation — is unchanged. Simulation groups should apply the same r/r_d stacking protocol with the same fitting methodology and the same Υ_* assumption when comparing to the parent paper's numerical benchmark.

Independent replication on THINGS, LITTLE THINGS, and MIGHTEE-HI (Section G), along with the radial $\Upsilon_*(r)$ re-analysis (Section D) and the Zhao $\alpha\beta\gamma$ profile scan (Section F), are the priority follow-up analyses for elevating the parent paper's finding from a SPARC-specific result to a population-level statement about late-type spiral kinematics.

Additional References

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